ARCHIT AGRAWAL

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PROFESSIONAL EXPERIENCE

EdPlus, Arizona State University **06/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Own architecture and delivery of an AI analytics platform on AWS Bedrock and Step Functions that turns any structured dataset into cited, code-computed answers.
- Validated 25 statistical tests and six causal estimators against SciPy and statsmodels; scored 68% on a 100-question StatQA sample, against GPT-4o's best reported 64.83%.
- Improved accuracy from 60% to 93% on the hardest benchmark questions by consolidating column definitions in a shared semantic layer.
- Cut incorrect answers 85% in an evaluation against 164 published benchmark figures by adding answer verification before display.
- Quadrupled the headline findings automated reports recover, 13% to 52–59% on held-out datasets, by letting an LLM choose analyses while code computes every number.
- Showed model choice moves labeling F1 33x more than pipeline design (0.230 vs 0.007), redirecting the team toward reliability.
- Designed a chart recommender covering 17 chart types, retaining deterministic rules when LLM proposals failed to improve evaluation scores.

MyStage Music Inc. **03/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Consolidated a 14-Cloud-Function backend into a checkpointed Python/LangGraph workflow on GCP Cloud Run, reducing production service dependencies 64%.
- Built FastAPI and Algolia search with sub-50 ms latency, real-time Firestore synchronization, 200+ pytest tests and Logfire tracing.
- Scaled ingestion to 70,000+ records daily from 650+ venues using Playwright, transactional locking and entity resolution.
- Implemented checkpointed pause-and-resume so records missing context wait for additional information without failing the ingestion batch.

EdPlus, Arizona State University **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Established Prompt Flow release checks for groundedness, relevance and coherence, evaluating prompt changes against measured results before deployment.
- Added hybrid keyword and vector retrieval to resolve exact course codes alongside semantic document search.
- Built a multi-tenant course assistant from React UI to retrieval services, used by 1,000+ faculty creating courses for 60,000+ students.
- Compared fixed-size and section-aware chunking on course documents to preserve complete policy passages in retrieved results.
- Reduced transcript-analysis time from four hours to 15 minutes using a Neo4j knowledge graph and validated Cypher queries.

Arizona State University **08/2024 - 05/2025**
Student Researcher *Phoenix, AZ*

- Built a PyTorch and Diffusers benchmark for text-to-video models that scores physical plausibility separately from visual quality, holding seed, resolution and sampler fixed.
- Found visual quality barely predicts physical plausibility: the top model by human preference ranked last on causal consistency.
- Designed a frame-level rubric for rigid-body motion, gravity, momentum and occlusion, and measured inter-rater agreement before trusting any aggregate score.
- Derived a failure taxonomy from annotated clips: broken object permanence, non-conserved mass, teleporting contact and physics that resets at a scene cut.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Reduced p95 retrieval latency 60% at 95% recall using hybrid FAISS and Neo4j search.
- Built a FastAPI and pgvector collaborator recommender achieving 85% top-k precision in evaluation.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Reduced deployment time 70% and production incidents 40% through containerized CI/CD and SonarQube quality gates.

DA-IICT **05/2021 - 08/2021**
Computer Vision Researcher *Gandhinagar, India*

- Engineered an autonomous-driving perception stack in Python: radar and lidar capture, timestamp alignment, extrinsic calibration and point-cloud processing, with a replay harness over recorded drives.
- Compared early and late sensor fusion on identical sequences, showing point-level merging keeps detail but inherits both sensors' noise while object-level merging is robust but discards evidence.

PROJECTS

AiJockey – AI DJ Pipeline | GitHub **2025 - Present**

- Built a PyTorch/FastAPI audio pipeline with stem separation, LLM transition planning and 25+ DSP transitions; ported inference to AMD MI300X.
- Trained a MERT-95M reward head to score audio segments across four aesthetic dimensions for candidate selection.
- Implemented DPO, KTO and IPO preference-training variants; reduced DPO training loss from 0.68 to 0.47 on labeled pairs.

TECHNICAL SKILLS

Languages & Models: Python, SQL, PyTorch, Diffusers, NumPy, SciPy, statsmodels
Retrieval & Evaluation: FAISS, pgvector, Neo4j, RAG, LLM evaluation, LangGraph
Serving & Cloud: FastAPI, AWS Bedrock, GCP Cloud Run, Docker, pytest

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
M.S., Computer Science | GPA: 4.0 *Tempe, AZ*

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
B.Tech., Information and Communication Technology *Gandhinagar, India*